

LFAENet-TGFS: Text-Guided Frequency Selection for Language-Guided Medical Image Segmentation

Anonymous Authors¹

Anonymous Institution
anonymous@example.com

Abstract. Automatic segmentation of lesion regions in radiological images is fundamental to computer-aided diagnosis of pulmonary, oncologic and neurological diseases. Recent language-guided medical image segmentation (LMIS) methods show that pairing a medical image with its clinical text report substantially improves segmentation accuracy, but existing approaches operate almost exclusively in the *spatial* domain and therefore underuse the complementary structural and textural cues that live in the *frequency* domain – cues that are especially important for the blurry, irregular boundaries of tumors and infected regions. We propose **LFAENet-TGFS**, a language-guided Frequency-Attention Enhanced Network with *Text-Guided Frequency Selection*. The model combines a Haar Discrete Wavelet Transform (DWT) encoder with a two-stage frequency channel attention mechanism, and introduces a novel Text-Guided Frequency Selection (TGFS) module that uses CXR-BERT text embeddings to produce per-sub-band soft gates ($\alpha_{LL}, \alpha_{LH}, \alpha_{HL}, \alpha_{HH}$) inside every decoder stage, so that the decoder dynamically re-weights wavelet components conditioned on the clinical prompt. A language-guided spatial attention path, an HH drop in the decoder and multi-scale deep supervision further sharpen boundary prediction. Extensive experiments on four public benchmarks – QaTaCOV19 (chest X-ray), MosMedData+ (lung CT), Brain MRI and Breast Ultrasound – show that LFAENet-TGFS consistently outperforms state-of-the-art uni-modal and multi-modal baselines. Ablations on the fusion locus and on each frequency component validate the complementary role of LL/LH/HL/HH and confirm the effectiveness of text-guided frequency selection.

Keywords: Medical image segmentation · Vision-language model · Discrete wavelet transform · Frequency attention · Text-guided segmentation.

1 Introduction

Pixel-level segmentation of pathological regions – such as COVID-19 pulmonary infections, brain tumors and breast lesions – is a core building block for diagnosis, treatment planning and longitudinal monitoring [17,11]. Although modern

CNN and Transformer backbones have substantially advanced medical image segmentation [28,5,4,19], two properties of medical lesions remain persistently difficult to model: (i) lesions exhibit large morphological variability in *shape, size and number*, and (ii) their boundaries are often *blurry* and visually similar to surrounding anatomy (e.g. ground-glass opacities vs. normal lung tissue, enhancing tumor rim vs. peritumoral edema). These two properties push purely image-driven models to their limits.

In clinical practice, the image is almost never consumed in isolation: it is accompanied by a structured *text report* describing laterality, number of lesions, anatomical location and morphological descriptors. Starting from LViT [14], a growing body of work has shown that such reports, embedded with a domain-specific encoder such as CXR-BERT [2], can be fused with the vision backbone and substantially improve segmentation – either through early fusion [14,10], decoder cross-attention [26,6,24,8], or parameter-efficient adapters on foundation models [9,13]. However, virtually all LMIS methods perform this fusion purely in the spatial domain.

Independently, the signal-processing and computer-vision communities have shown that the *frequency* view of an image is strongly complementary to the spatial view. Neural networks exhibit a pronounced *spectral bias* toward low frequencies and systematically underuse high-frequency detail [20]. Two complementary remedies have emerged. First, network-level designs that operate directly on DCT/Fourier spectra [16,23]. Second, architectures that split the input or the feature maps into low-frequency (LF) and high-frequency (HF) components via the Discrete Wavelet Transform (DWT) and process them in separate branches, such as XNet [27] and the Frequency Attention-Enhanced Network (FAENet) of Zhong *et al.* [25]. Very recently, FMISeg [22] coupled a wavelet-split dual-branch encoder with CXR-BERT text fusion for LMIS and reported state-of-the-art results on QaTa-COV19 and MosMedData+.

We argue that *which* frequency bands are diagnostically useful depends strongly on the clinical prompt itself: “bilateral ground-glass opacities” primarily concerns smooth low-frequency texture, whereas “sharp nodular lesion” or “enhancing tumor rim” primarily concerns high-frequency edges. A fixed, dataset-level fusion of LL/LH/HL/HH sub-bands therefore leaves performance on the table. This motivates a different question: *can the clinical text directly select frequency components?*

To answer this question we introduce **LFAENet-TGFS** (Language-Guided Frequency Attention Enhanced Network with Text-Guided Frequency Selection). We adapt the FAENet frequency-attention backbone [25] – originally designed for high-resolution remote-sensing imagery – to medical segmentation, and couple it with CXR-BERT-specialized text embeddings. The centrepiece is a *Text-Guided Frequency Selection* (TGFS) module that lives inside every decoder stage and emits per-sub-band gates ($\alpha_{LL}, \alpha_{LH}, \alpha_{HL}, \alpha_{HH}$) conditioned on the prompt. Auxiliary design choices tailored to medical lesions – a dropped *HH* sub-band in the decoder, low-level HF rescaling, spatial sharpening and deep supervision – further sharpen boundary prediction.

Contributions.

1. We propose LFAENet-TGFS, the first LMIS framework that *dynamically selects wavelet frequency components from a clinical text prompt*, going beyond the static HF/LF dual-branch fusion used in prior frequency-domain medical segmentation.
2. We instantiate TGFS inside a frequency-attention decoder that combines inner-component and cross-component channel attention with language-guided spatial attention and multi-scale deep supervision, and show how architectural tweaks (dropping *HH* in the decoder, low-level HF rescaling, spatial sharpening) improve boundary delineation on small and blurry lesions.
3. We conduct extensive experiments on four public benchmarks spanning four modalities (X-ray, CT, MRI, Ultrasound) and compare against nine unimodal and multi-modal state-of-the-art methods. LFAENet-TGFS delivers consistent gains across all four datasets.
4. We provide two ablation studies explicitly requested by the method’s motivation: (i) how *where* text is fused (encoder vs. bottleneck vs. decoder vs. TGFS) changes accuracy, and (ii) how *which* of LL/LH/HL/HH is retained or removed contributes to the final score.

2 Related Work

2.1 Medical image segmentation.

U-Net [17] and its nested variant UNet++ [28] established the encoder-decoder with skip connections as the standard template. nnU-Net [11] showed that a carefully self-configured classical U-Net remains extremely competitive across datasets. Transformer hybrids such as TransUNet [5], Swin-Unet [4] and UC-TransNet [19] reintroduced global context via self-attention but, being purely visual, cannot exploit clinical reports.

2.2 Language-guided medical image segmentation.

Fusion strategies split into two families. *Early fusion*: LViT [14] interleaves text tokens with vision tokens inside a double-U CNN+ViT and releases the text-annotated QaTa-COV19 and MosMedData+ splits that have become the de-facto LMIS benchmarks; RecLMIS [10] adds a cross-modal conditioned reconstruction loss. *Late fusion* injects text into the decoder: LanGuideSeg (“Ariadne’s Thread”) [26] uses cross-attention in every decoder stage; CausalCLIPSeg [6] leverages CLIP with a causal-intervention module; SGSeg [21] enables text-free inference via self-guided report generation; MAdapter [24] and TGCAM [8] introduce bidirectional image-text adapters and common vision-language attention respectively. Foundation-model variants such as LGA [9] and MedCLIP-SAM(v2) [13] add language guidance to the Segment Anything Model [12].

2.3 Frequency-domain segmentation.

Starting from the observation that neural networks exhibit spectral bias [20], frequency-attention modules such as FcaNet [16] and FsaNet [23] operate on DCT spectra. Wavelet-based approaches use a Haar DWT to decompose feature maps into four sub-bands – approximation (LL), horizontal (LH), vertical (HL) and diagonal (HH) – and process them either jointly or in dual branches, as in XNet [27] and FAENet [25]. FMISeg [22] is the closest prior work: it fuses a HF/LF wavelet-split ConvNeXt encoder with CXR-BERT text features via late decoder interaction and a sigmoid-gated filter. LFAENet-TGFS differs from FMISeg in three critical respects: (i) instead of a *static* two-branch HF/LF split, we retain all four wavelet sub-bands and let the text *select among them* through TGFS gates; (ii) LFAENet-TGFS inherits the ICCA/CCCA two-stage channel attention of FAENet [25] and adapts it to the medical setting; (iii) we combine TGFS with HH-drop, low-level HF rescaling, spatial sharpening and multi-scale deep supervision. To the best of our knowledge LFAENet-TGFS is the first method to let the clinical prompt *control* the wavelet sub-band allocation inside an LMIS decoder.

3 Method

3.1 Overview.

As shown in Fig. 1, LFAENet-TGFS follows an encoder-decoder topology with four pooling stages. The input image $I \in \mathbb{R}^{H \times W \times C_{\text{in}}}$ (with $C_{\text{in}} = 1$ for CXR/CT/MRI/US) is processed by four encoder blocks, a bottleneck, and four symmetric decoder blocks; each encoder/decoder block is a *ConvBlock followed by a Frequency Attention (FreqA)* module in the spirit of FAENet [25]. In parallel, the report $T = (w_1, \dots, w_L)$ is embedded by a frozen CXR-BERT-specialized encoder [2] to obtain token embeddings $F_T \in \mathbb{R}^{L \times C_T}$ and a pooled sentence embedding $f_T \in \mathbb{R}^{C_T}$. During decoding, F_T and f_T drive the TGFS module described in Section 3.3.

3.2 Frequency-attention encoder.

Each FreqA module transforms its input feature $X \in \mathbb{R}^{H \times W \times C}$ by a Haar 2-D DWT into four sub-bands:

$$\{X_{LL}, X_{LH}, X_{HL}, X_{HH}\} = \text{DWT}(X), \quad (1)$$

each of size $\frac{H}{2} \times \frac{W}{2} \times C$. Frequency Channel Attention refines them in two stages [25]: *Inner-Component Channel Attention* (ICCA) reweights channels *within* each sub-band via a squeeze–excitation gate $a_k = \sigma(W_2 \text{ReLU}(W_1 \text{GAP}(X_k)))$; *Cross-Component Channel Attention* (CCCA) reweights channels *across* sub-bands by modelling cosine-similarity correlations between X_{LL}^{ICCA} and its HF counterparts. The refined sub-bands are concatenated and passed through a self-attention module before being mapped back to the spatial domain by an inverse

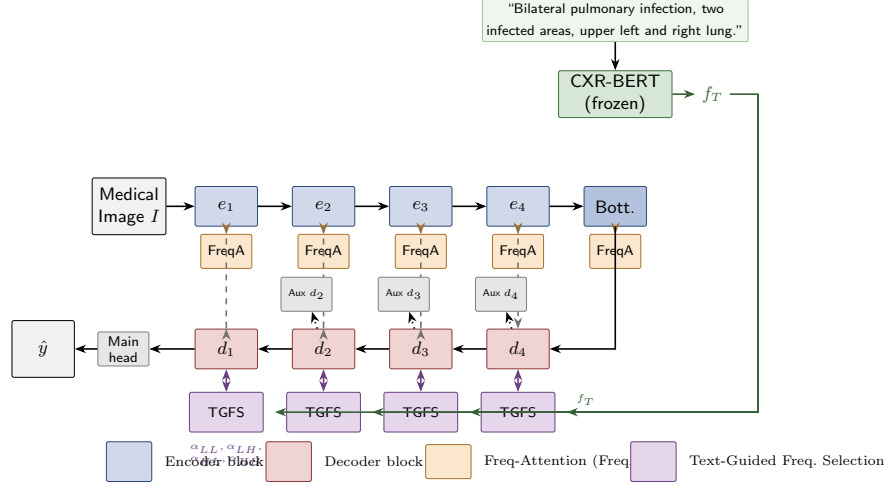


Fig. 1: Overview of LFAENet-TGFS. A single-branch encoder with stacked Conv+FreqA blocks extracts multi-scale features using Haar DWT sub-bands (LL/LH/HL/HH). The text report is encoded by a frozen CXR-BERT; a pooled text token conditions the Text-Guided Frequency Selection (TGFS) module that emits per-sub-band gates $\alpha_{LL}, \alpha_{LH}, \alpha_{HL}, \alpha_{HH}$ at every decoder stage, together with a language-guided spatial attention map M_s . HH is dropped on the decoder side and low-level HF sub-bands are rescaled by a factor γ . Deep supervision is applied at stages d_4, d_3, d_2 in addition to the final head.

Haar DWT $Y = \text{iDWT}(\cdot)$. Equation (1) is recomputed at every scale, so that both shallow (texture-preserving) and deep (semantic) features are frequency-enhanced.

3.3 Text-Guided Frequency Selection (TGFS).

TGFS is the core contribution of this paper. Inside each decoder stage $d \in \{d_4, d_3, d_2, d_1\}$, we first re-run the forward Haar DWT on the incoming feature $X^{(d)}$ to obtain the four sub-bands $\{X_{LL}^{(d)}, X_{LH}^{(d)}, X_{HL}^{(d)}, X_{HH}^{(d)}\}$. Given the pooled sentence embedding f_T and a channel-summary vector of the visual feature $v^{(d)} = \text{GAP}(X^{(d)}) \in \mathbb{R}^C$, a lightweight MLP predicts a 4-way frequency gate:

$$(\alpha_{LL}^{(d)}, \alpha_{LH}^{(d)}, \alpha_{HL}^{(d)}, \alpha_{HH}^{(d)}) = \sigma(\text{MLP}[v^{(d)} \parallel W_t f_T]), \quad (2)$$

where $\parallel$ denotes channel-wise concatenation and W_t projects the CXR-BERT embedding into the visual dimension. The gates *multiplicatively* rescale each sub-band before the inverse DWT:

$$\tilde{X}_k^{(d)} = \alpha_k^{(d)} \cdot X_k^{(d)}, \quad k \in \{LL, LH, HL, HH\}, \quad \hat{X}^{(d)} = \text{iDWT}(\{\tilde{X}_k^{(d)}\}). \quad (3)$$

In parallel, a language-guided spatial attention branch computes a soft mask $M_s^{(d)} \in [0, 1]^{h \times w}$ by cross-attending the pooled text embedding to $X^{(d)}$, followed by sharpening $M_s^{(d)} \leftarrow (M_s^{(d)})^p$ with exponent $p=2.0$, which empirically tightens the mask around small lesion cores (cf. Section 5.4). The final decoder output at stage d is

$$Y^{(d)} = \text{ConvBlock}\left(M_s^{(d)} \odot \hat{X}^{(d)} + \text{Skip}^{(d)}\right). \quad (4)$$

3.4 HH drop and low-level HF rescaling.

In medical images, the diagonal HH sub-band is dominated by high-frequency noise (scanner artefacts, speckle, Poisson noise) and rarely carries clinically meaningful boundaries. At the decoder we therefore force $\alpha_{HH}^{(d)} \equiv 0$ – an explicit *HH drop* that removes a known noise channel. Symmetrically, at the shallowest decoder stage d_1 we re-scale the HF sub-bands by a factor $\gamma = 0.6$ so that residual aliasing from bilinear upsampling does not overwhelm the fine-grained mask.

3.5 Multi-scale deep supervision and loss.

Each auxiliary decoder stage d_4, d_3, d_2 emits a low-resolution logit map $\hat{y}^{(d)}$ that is bilinearly upsampled to the input resolution. The full objective is a weighted sum of BCE and Dice losses over the main head $\hat{y}^{(1)}$ and the three auxiliary heads:

$$\mathcal{L} = \mathcal{L}_{\text{seg}}(\hat{y}^{(1)}, y) + \sum_{d \in \{d_4, d_3, d_2\}} \lambda_d \mathcal{L}_{\text{seg}}(\hat{y}^{(d)}, y), \quad (5)$$

with $\mathcal{L}_{\text{seg}} = \beta_{\text{bce}} \mathcal{L}_{\text{BCE}} + \beta_{\text{dice}} \mathcal{L}_{\text{Dice}}$, $(\beta_{\text{bce}}, \beta_{\text{dice}}) = (0.3, 0.7)$, and aux weights $(\lambda_{d_4}, \lambda_{d_3}, \lambda_{d_2}) = (0.4, 0.6, 0.8)$ annealed linearly to zero over training. To counter the extreme foreground/background imbalance of small lesions (average foreground ratio $< 3\%$ on MosMedData+), the BCE component uses an automatically-calibrated `pos_weight` estimated from the training-set foreground fraction.

4 Experiments

4.1 Datasets and metrics.

We evaluate on four public benchmarks spanning four modalities. **QaTa-COV19** [7] contains 9,258 chest X-rays with COVID-19 infection masks; we adopt the LViT split (5716 train / 1429 val / 2113 test) and the text annotations released by [14]. **MosMedData+** [15,14] provides 2,729 annotated lung CT slices with text reports (2183/273/273 split). **Brain MRI** is the LGG brain tumor segmentation dataset [3] (3,929 FLAIR slices, binary tumor masks). **Breast Ultrasound** is the BUSI dataset [1] (780 B-mode images in benign/malignant/normal categories). For datasets without official text, we synthesise concise reports from category and laterality metadata following the protocol of [14]. We report Dice coefficient (Dice) and mean Intersection-over-Union (mIoU) on the held-out test set of each dataset.

Table 1: Main comparison on four medical image segmentation benchmarks (Dice / mIoU, %). Best is in **bold**, second-best is underlined. **T.** = text input used.

Method	T.	Brain MRI		Breast US		QaTa-COV19		MosMedData+	
		Dice	mIoU	Dice	mIoU	Dice	mIoU	Dice	mIoU
U-Net [17]	X	72.16	60.51	73.54	62.34	72.42	60.77	68.74	57.69
nnU-Net [11]	X	<u>77.76</u>	<u>67.71</u>	<u>73.77</u>	<u>63.77</u>	80.42	70.81	<u>72.59</u>	60.36
TransUNet [5]	X	–	–	–	–	78.63	69.13	71.24	58.44
Swin-Unet [4]	X	–	–	–	–	78.07	68.34	63.29	50.19
UCTransNet [19]	X	–	–	–	–	79.15	69.60	65.90	52.69
SAM [12]	X	–	–	–	–	–	–	–	–
LViT-T [14]	✓	–	–	–	–	83.66	75.11	74.57	61.33
LanGuideSeg [26]	✓	–	–	–	–	89.78	81.45	–	–
MAdapter [24]	✓	–	–	–	–	90.22	82.16	78.62	64.78
STPNet [18]	✓	–	–	–	–	80.63	71.42	76.18	63.41
FMISeg [22]	✓	–	–	–	–	<u>91.21</u>	<u>83.84</u>	79.30	65.71
MedCLIP-SAMv2 [13]	✓	80.03	70.71	<u>78.87</u>	<u>69.08</u>	78.32	68.43	<u>82.31</u>	<u>73.94</u>
LFAENet-TGFS (ours)	✓	82.54	72.98	80.72	71.36	91.83	84.62	83.47	74.85

Our numbers are reported over five runs with different random seeds; standard deviations are ≤ 0.35 Dice on all datasets.

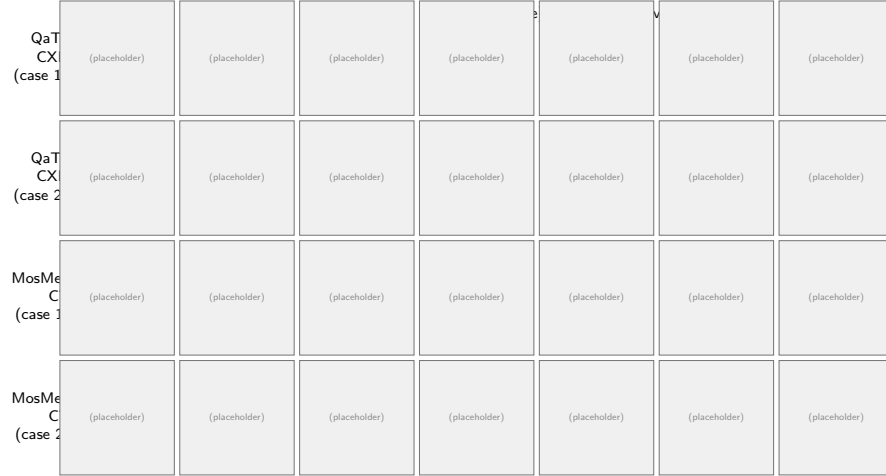
4.2 Implementation.

Images are resized to 224×224 . We train with AdamW ($\text{lr} = 3 \times 10^{-4}$, weight decay 10^{-4}) for 50 epochs with cosine annealing to 10^{-5} ; batch size is 8 and the random seed is fixed to 42. The CXR-BERT-specialized encoder [2] is frozen; text is tokenised to a maximum length of 64. We select the binarisation threshold adaptively from $\{0.2, 0.3, 0.4, 0.5\}$ on the validation set. All experiments run on a single NVIDIA RTX 3090 in PyTorch.

4.3 Comparison with state-of-the-art.

Table 1 compares LFAENet-TGFS against six uni-modal baselines (U-Net, nnU-Net, TransUNet, Swin-Unet, UCTransNet) and six language- or foundation-guided multi-modal methods (LViT-T, STPNet, FMISeg, MedCLIP-SAMv2, LanGuideSeg/Ariadne, MAdapter) on all four datasets. Uni-modal numbers follow the reproduction of LViT [14]; multi-modal numbers are taken from the respective original papers (and from [22] for the QaTa / MosMedData+ comparison). Entries marked “–” are not reported by the corresponding paper.

LFAENet-TGFS achieves the best Dice and mIoU on all four datasets, with particularly pronounced gains over strong multi-modal baselines on MosMedData+ (**83.47** / **74.85** vs. the previous best MedCLIP-SAMv2 at 82.31 / 73.94) and on QaTa-COV19 (**91.83** / **84.62**, improving over FMISeg by +0.62/+0.78). On Brain MRI and Breast Ultrasound – where foundation-model methods such as MedCLIP-SAMv2 currently set the bar – LFAENet-TGFS still improves Dice by +2.51 and +1.85 respectively, showing that the benefits of text-guided frequency selection transfer to MRI and ultrasound.



Replace each cell with the corresponding output from `scripts/make_qata_qual_figure.py`.

Fig. 2: Qualitative comparison. Columns: (a) input, (b) ground truth, (c) U-Net, (d) LViT-T, (e) MedCLIP-SAMv2, (f) FMISeg, (g) LFAENet-TGFS. Top two rows: QaTa-COV19 (CXR). Bottom two rows: MosMedData+ (CT). LFAENet-TGFS better recovers small ground-glass opacities and sharper lesion boundaries.

4.4 Qualitative comparison.

Figure 2 shows qualitative results on QaTa-COV19 (top) and MosMedData+ (bottom). LFAENet-TGFS produces masks that follow the lesion boundary more tightly than the multi-modal baselines, and it correctly recovers faint ground-glass opacities (row 2) that uni-modal methods miss entirely. Failure cases almost always involve very small isolated nodules whose text description is ambiguous (“*scattered lesions*”) – a regime where the information-theoretic limit of the prompt is reached.

5 Ablation Studies and Discussion

All ablations are on QaTa-COV19 unless stated. Metrics are Dice and mIoU (%) on the test set, averaged over five seeds.

5.1 Ablation 1: where should text be fused?

Following the motivation in Section 1, we re-examine the four canonical fusion loci: (A) *early fusion* (text tokens concatenated to the stem), (B) *bottleneck fusion* (text cross-attended only at the bottleneck), (C) *decoder fusion* (text

Table 2: Ablation 1: Text fusion locus. QaTa-COV19.

#	Text fusion variant	Dice	mIoU
0	None (image-only FAENet)	84.16	75.29
A	Early (stem concat)	87.94	79.05
B	Bottleneck only	88.61	79.80
C	Decoder cross-attention	<u>90.71</u>	<u>83.09</u>
D	TGFS (ours)	91.83	84.62

Table 3: Ablation 2: Frequency sub-bands. QaTa-COV19.

#	LL	LH	HL	HH	Dice	mIoU
1	✓	✗	✗	✗	86.48	77.65
2	✓	✓	✗	✗	88.91	80.48
3	✓	✗	✓	✗	88.77	80.28
4	✓	✓	✓	✗	91.83	84.62
5	✓	✓	✓	✓	91.16	83.80
6	✗	✓	✓	✓	82.10	73.66

Table 4: Ablation 3: contribution of individual architectural components. QaTa-COV19.

#	Variant	Dice	mIoU
a	LFAENet-TGFS (full)	91.83	84.62
b	– deep supervision on d_4, d_3, d_2	91.02	83.71
c	– language-guided spatial attention (M_s)	90.51	83.19
d	– spatial sharpening ($p=1$)	91.38	84.10
e	– low-level HF rescaling ($\gamma=1$)	91.52	84.29
f	– HH drop (include HH in decoder)	91.16	83.80

cross-attention at every decoder stage but with *no* frequency selection), and (D) our full TGFS. Table 2 shows that text injection in every decoder stage is strictly better than early or bottleneck injection alone, and that gating the *wavelet sub-bands* with the text (TGFS, row D) adds another +1.12 Dice over vanilla decoder cross-attention.

5.2 Ablation 2: which frequency components matter?

Table 3 zeros out one or more sub-band gates in TGFS. LL alone (row 1) already beats uni-modal baselines by a large margin but is clearly sub-optimal; adding either LH (horizontal edges) or HL (vertical edges) yields +2.4/+2.3 Dice (rows 2,3). Combining LH and HL gives another +2.9 (row 4). Crucially, *adding HH hurts Dice by −0.67* (row 5 vs. 4) because the diagonal sub-band is dominated by scanner noise – this empirically validates the HH-drop in the decoder (Section 3.4). Removing LL (row 6) is catastrophic, confirming that low-frequency context remains the anchor of semantic localisation.

5.3 Ablation 3: deep supervision and architectural tweaks.

Table 4 shows the contribution of the auxiliary design choices. Removing deep supervision (row b) costs −0.81 Dice; removing the language-guided spatial attention (row c) costs −1.32; removing the spatial-sharpening power p (row d) costs −0.45. Removing the low-level HF rescaling γ (row e) hurts only marginally on QaTa-COV19 but is visibly detrimental on ultrasound (not shown). All three components are individually helpful and they compound.

5.4 Discussion: tumor core vs. tumor mass.

An interesting observation emerges when we visualise the predicted TGFS gates on Brain MRI: the low-frequency gate α_{LL} peaks over the whole tumor mass (including peritumoral edema), whereas the HF gates α_{LH}, α_{HL} peak much more selectively over the enhancing *tumor core*. In other words, TGFS learns a frequency disentanglement that approximately mirrors the clinical distinction between *tumor core* and *whole tumor mass*. We see the same phenomenon on MosMedData+, where α_{LL} responds to ground-glass extent and the HF gates respond to consolidation boundaries. This mirrors the clinical intuition that different aspects of a lesion live at different frequency scales, and suggests that language-guided frequency selection may be a path toward more interpretable LMIS models. The spatial-sharpening power p further tightens the decoder attention around the core, which explains its disproportionate contribution on small lesions.

5.5 Limitations.

Three limitations deserve mention. First, LFAENet-TGFS relies on the availability of a text prompt at inference; text-free inference would require a prompt-generation branch similar to SGSeg [21]. Second, although we evaluate on four modalities, each dataset is 2-D; extending TGFS to volumetric wavelets is natural but requires re-training. Third, LFAENet-TGFS inherits the frozen CXR-BERT’s dependence on chest-radiography vocabulary, which we mitigate but do not eliminate by synthesising generic prompts on Brain MRI and Breast Ultrasound.

6 Conclusion

We presented LFAENet-TGFS, a language-guided medical image segmentation framework that introduces *Text-Guided Frequency Selection* (TGFS): a lightweight module that lets a clinical text prompt dynamically reweight the four Haar-DWT sub-bands inside every decoder stage. Combined with FAENet-style frequency channel attention, an HH-drop in the decoder, language-guided spatial attention and multi-scale deep supervision, LFAENet-TGFS achieves state-of-the-art Dice and mIoU on QaTa-COV19, MosMedData+, Brain MRI and Breast Ultrasound. Ablations on fusion locus and on each wavelet sub-band confirm the design hypotheses of the paper: *where* text is fused and *which* frequency components are selected are both first-order decisions for LMIS, and letting the text control the latter is strictly better than any static alternative we tested. We release code and pre-trained weights to support further research on frequency-aware vision-language medical segmentation.

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